Open and Challenging Problems in Reconfigurable, Metamorphic, Foldable, Origami, and Kirigami Mechanisms

Executive Summary

Origami/kirigami and related reconfigurable (including metamorphic) mechanisms now span **discrete** systems (crease patterns, linkage-and-panel networks, modular foldable robots) and **continuum** systems (elastic sheets, kirigami cuts, architected lattices, thick-panel deployables). Reviews over the past decade emphasize rapid growth in (i) **mechanics and simulation**, (ii) **inverse design / synthesis**, and (iii) **applications** ranging from robotics and aerospace deployables to electronics and electromagnetic structures. [1]

Despite this maturity, several core problems remain open because they combine: **high-dimensional constrained kinematics**, **contact and self-avoidance**, **instabilities and multistability**, **material/thickness effects**, and **multi-physics actuation–sensing–control co-design**. These difficulties are not merely engineering nuisances; foundational results show that even *idealized* decision problems like rigid foldability can be computationally intractable (NP-hard), which constrains what "fully general" algorithms can guarantee. [2]

This report highlights **eight** unsolved problems that cut across theory, synthesis, and practice. For each, it states (a) the problem, (b) why it remains unsolved, (c) prior approaches and limitations, and (d) concrete technical routines that could plausibly move the field forward, with estimated difficulty, expertise, and timelines.

A unifying theme emerges: near-term progress is most plausible when research shifts from "solve everything exactly" to **hierarchical pipelines** that couple (1) **topology/rigidity screening**, (2) **reduced-order differentiable simulation with contact**, (3) **robust optimization or learning-based inverse design**, and (4) **application-specific verification protocols** (fatigue, environmental cycling, calibration/benchmarks). [3]

Scope and Technical Taxonomy

Origami/kirigami mechanisms can be organized along four axes that matter for "open problems":

Discrete vs continuum.
Discrete models treat panels as rigid bodies connected by hinges (creases) and sometimes springs (bar-and-hinge, linkage networks). *Continuum* models treat sheets and lattices as elastic shells/solids where bending, stretching, buckling, and

cut-induced mechanisms all contribute. Reviews of origami simulations and origami/kirigami metamaterials emphasize that most practical systems are **hybrid**: they rely on discrete kinematics but are controlled/limited by continuum mechanics and contact. [4]

Rigid vs deformable folding.
Rigid origami is an idealization (panels rigid; motion only at creases). Deformable origami/kirigami stores energy in both creases and panels, producing richer (and harder) energy landscapes and stability phenomena. [5]

Single-mode vs multi-mode / metamorphic.
Some designs aim for a single deployment path; others exploit bifurcations and multistability, or deliberately change effective constraints/topology (metamorphic behaviors) to access many configurations. Recent work explicitly leverages bifurcation for reconfigurable metamaterials, while other work builds “lockable” flat-foldable but load-bearing states via contact. [6]

Single-physics vs multi-physics.
Common actuation spans thermal/electrothermal, pneumatic/vacuum, magnetic, shape-memory, motors/cables, etc. Multi-physics origami systems and field-programmable folding sheets illustrate that *actuation and sensing hardware* increasingly dominate what can be achieved in real time. [7]

Assumptions used in this report

Because no constraints were specified, this report assumes: - **Scale** is unconstrained (micro/nano sheets and mesoscale devices up to meter-scale deployables). Mechanically guided 2D→3D assembly demonstrates feasibility from micro to macro scales, but also shows that dominant phenomena shift with thickness, fabrication tolerance, and material nonlinearity. [8] - **Materials** are unconstrained (polymers, metals, composites, paper-like laminates, thin films, multi-material prints, shape memory polymers). - **Actuation modes** include thermal/electrothermal, pneumatic/vacuum, magnetic, shape-memory, and motors/cables, with implications discussed where relevant. [9]

Theoretical Analysis Methods: Core Unsolved Problems

Global configuration-space characterization and fold-path planning with self-contact and branching

Problem statement (a).
Develop a **general, computationally tractable** method to:
1) determine whether an origami/kirigami (or panel-hinge / linkage-sheet hybrid) system is physically foldable/deployable from a given start configuration,
2) compute **at least one** feasible folding path to a target configuration, and

3) do so while enforcing **non-interpenetration**, handling **self-contact**, and reliably managing **bifurcations/multiple folding modes**.

This problem spans rigid-origami kinematics, thick-panel designs, kirigami cuts that create new mechanism DOFs, and metamorphic transitions created by contact/locking. [10]

Why it is unsolved (b): key technical challenges.
- **Computational hardness**: deciding rigid foldability is NP-hard in general, which strongly suggests that fully general polynomial-time algorithms with guarantees are unlikely without restricting problem structure. [11]
- **High-dimensional, constrained motion**: rigid origami uses nonlinear compatibility constraints on folding angles; local rigidity analysis requires Jacobians/Hessians with higher-order structure compared with typical bar-joint frameworks. [12]
- **Self-contact and stacking**: contact can *create* rigidity/locking and prune motion branches, but classic local algebraic analyses often omit self-intersection/stacking sequence effects explicitly (they are flagged as needing further work). [13]
- **Bifurcations and "distractor" branches**: even patterns intended to have a single folding motion can exhibit many accessible branches near the flat state, complicating actuation planning and path selection. [14]

Prior approaches and limitations (c).
- **Local rigidity / differential analysis** has advanced (e.g., first/second-order rigidity definitions for rigid origami), but its scope is primarily local and does not resolve global self-contact/stacking or global motion planning. [15]
- **Explicit kinematics for certain vertices** (e.g., degree-4 rigid origami vertices) improves analytic tractability but does not scale to large patterns with contact and heterogeneous panel geometries. [16]
- **Contact simulation frameworks** based on energy methods and barrier-like contact potentials have been introduced to simulate panel contact efficiently, but robustly embedding these into global path planning and inverse design (especially with bifurcations) remains difficult. [17]
- **Energy-path methods** (e.g., nudged elastic band to find minimum-energy folding paths in multistable origami) demonstrate feasibility for some systems, but scaling and guaranteeing global feasibility across modes remains open. [18]

Concrete technical routines to address the problem (d).

A plausible research program is to treat fold-path planning as a **contact-constrained continuation + motion planning** pipeline:

1) **Model unification (hybrid coordinates).**
Represent the system with generalized coordinates q that include: - fold angles (hinge DOFs),
- optional panel bending modes (reduced basis),

- and, for kirigami, mechanism DOFs induced by cuts.
Impose constraints $g(q) = 0$ (kinematic compatibility) and inequalities $d_i(q) \geq 0$ (signed distances for nonpenetration).

2) **Differentiable contact barrier.**
Adopt a barrier potential approach for contact where $U_{\text{contact}}(q) \to \infty$ as $d_i(q) \to 0^+$, consistent with energy-based contact potentials used in origami contact simulation. [19]
Then define a total potential $U(q) = U_{\text{elastic}}(q) + U_{\text{contact}}(q)$ and search for low-energy feasible paths.

3) **Pseudo-arclength continuation with bifurcation detection.**
Use numerical continuation to trace solution branches of $g(q) = 0$ (or stationarity $\nabla U(q) = 0$ for quasi-static equilibria), with: - rank tests on Jacobians to detect near-singularities/bifurcations, consistent with the broader rigid-origami local analysis emphasis on Jacobian structure. [20]

4) **Branch management via "mode charts."**
At detected branch points, compute several candidate directions via nullspaces and follow each branch for a bounded arc-length, producing a graph of feasible mode segments.

5) **Sampling-based planning in "mode graph" space.**
Run PRM/RRT-style planning where nodes are equilibria on branches and edges are validated continuation segments; collision/contact is enforced by the barrier and/or explicit distance checks.

6) **Certified subproblems.**
Because NP-hardness blocks universal guarantees, focus on certifiable subclasses: - periodic tessellations with limited vertex types,
- thick-panel patterns satisfying known uniform-thickness conditions,
- or single-DOF rigidly foldable classes (e.g., designs intended to have one DOF plus contact-based locks). [21]

7) **Validation protocol.**
For each computed path, validate via: - physical prototypes (laser-cut or 3D-printed) with motion capture of fold angles,
- force–displacement curves, and
- contact event sequencing, comparing against the simulated times/angles of first contact and lock. Contact-heavy load-bearing origami metamaterial states make such validation essential. [22]

Difficulty, expertise, timeline.
- Difficulty: **Very high** (hardness results + contact + bifurcations). [23]
- Required expertise: computational geometry + nonlinear mechanics + numerical continuation + contact mechanics + robotics motion planning.

- Plausible timeline: - **1–2 years**: robust pipelines for restricted families (single-DOF + contact locks; selected tessellations). [22]
- **3–5 years**: mode-graph planners for moderate-sized patterns with validated contact.
- **5–10+ years**: broadly general toolchains with reliable performance across large heterogeneous systems (still unlikely to be "universal" due to NP-hardness). [24]

Unified, thickness-aware mechanics with error-controlled reduced-order models

Problem statement (a).
Create mechanics models that **bridge ideal crease kinematics** (zero-thickness rigid panels) with **real, thick, deformable, and load-bearing** systems, while providing: - predictable errors (bounds or at least calibrated uncertainty),
- tractable reduced-order behavior for simulation and design, and
- validity across the spectrum from thin films (microfabricated kirigami) to meter-scale thick-panel deployables.

Why it is unsolved (b): key technical challenges.
- Thickness introduces geometric incompatibilities and contact/groove issues, requiring modified crease-panel geometries and additional compatibility conditions. [25]
- Practical thick-panel systems must support load near deployed configurations; thin-panel kinematic models often fail precisely where load-bearing matters. [26]
- Reduced-order models (bar-and-hinge) can miss panel buckling, new creases, and real contact behavior, which are noted limitations in efforts to automate modeling. [27]

Prior approaches and limitations (c).
- Thick-panel origami geometry frameworks and demonstrations show improved load-bearing near deployment (e.g., thick Yoshimura studies), but generalizing to arbitrary crease networks and integrating with inverse design remains incomplete. [28]
- Large-scale modular uniformly thick origami-inspired structures incorporate thick-panel connectors and bar-and-hinge formulations to simulate deployment and load-carrying, with open-source code releases that improve reproducibility. However, general uncertainty quantification and coupling to fatigue/damage models remain open. [29]
- Reviews of origami simulations highlight many modeling approaches (kinematics → mechanics → multi-physics), but also underscore gaps in handling contact, thickness, and material realism simultaneously at scale. [30]

Concrete technical routines to address the problem (d).

1) **Hierarchical "kinematics-first, mechanics-refine" modeling.**
- Stage 1: solve an idealized rigid-folding kinematics for candidate configurations (or

a low-DOF parameterization).
- Stage 2: lift the solution into a thick-panel representation with explicit connector/hinge geometry.
- Stage 3: refine with a shell/solid FE submodel *only* where strain localizes (creases, connectors), enabling domain decomposition.

2) **Crease constitutive identification library.**
Build standardized hinge models (torsional springs, plastic hinges, viscoelastic hinges) calibrated from coupon tests, then propagate these into bar-and-hinge global simulations—consistent with approaches that separate crease and panel roles in origami structures. [31]

3) **Error-aware reduced bases.**
Use proper orthogonal decomposition / reduced basis methods on high-fidelity simulations of representative unit cells (origami/kirigami motifs), then enforce compatibility/contact constraints in the reduced coordinates.

4) **Homogenization with configuration dependence.**
For metamaterial lattices, derive effective stiffness or compliance tensors as functions of configuration parameters (fold angle, cut stretch). Recent kirigami and origami metamaterial work emphasizes programmable stiffness/constitutive behavior; the open step is making those mappings reliable across manufacturing variability. [32]

5) **Validation protocol.**
- Static: compare predicted vs measured load–displacement, stiffness near deployed states, and locking loads for contact-enabled stiff states. [33]
- Geometric: compare 3D shapes (photogrammetry/structured light) and fold-angle distributions against simulations at multiple configurations.
- Sensitivity: perturb thickness, hinge stiffness, and friction to quantify robustness envelopes.

Difficulty, expertise, timeline.
- Difficulty: **High** (multi-scale + contact + nonlinear materials).
- Expertise: nonlinear mechanics, shell/solid FE, reduced-order modeling, experimental mechanics, metrology.
- Timeline: - **1–3 years**: well-validated reduced-order + thick-panel pipelines for selected families (uniform-thickness, modular systems). [34]
- **3–6 years**: configuration-dependent homogenization + uncertainty quantification commonly usable for design.
- **6–10 years**: broadly transferable error-controlled models spanning micro-to-macro.

Energy landscape prediction and multistability/locking programmability under realistic loads

Problem statement (a).
Predict and design the **full energy landscape** (stable equilibria, energy barriers, transition paths) for origami/kirigami systems that exhibit: - multistability (bi-/tri-/multi-stable states),
- snapping and dynamic transitions,
- contact-induced locking (mechanical intelligence), and
- coupling to multi-physics actuation.

Why it is unsolved (b): key technical challenges.
- Landscapes are **rugged and branch-rich**: even "single-motion" patterns can have exponentially many distractor branches near flat states, making actuation planning and design for reliable folding difficult. [14]
- Multistability often depends on nonlinearities from panel bending, crease stiffness, and contact, so purely kinematic design rules are insufficient. [35]
- Energy barriers and transition paths must be computed in very high-dimensional spaces, where standard local solvers miss relevant saddles and bifurcations.

Prior approaches and limitations (c).
- Minimum-energy path computation (e.g., nudged elastic band) has been applied to multistable origami to find low-energy folding paths, but generalizing to contact-rich systems and scaling to large lattices is still emerging. [18]
- Lockable flat-foldable metamaterials show how contact can create stiff load-bearing states and multiple reconfigurable paths, but predicting *which* locks occur under realistic manufacturing tolerances remains challenging. [36]
- Kirigami metamaterials with programmable constitutive relations and deformation-dependent stiffness (including contact-induced stiffening) show promising deterministic geometry→mechanics mappings, but scaling these mappings to complex patterns with robust multistability is not solved. [37]

Concrete technical routines to address the problem (d).

1) **Landscape atlas construction.**
For a given design family, build an "atlas" of equilibria and transition states by combining: - continuation to trace equilibrium branches, and
- saddle-search methods (dimer method, NEB/string methods) to locate transition pathways between stable states. [18]

2) **Contact-aware barrier saddle search.**
Incorporate contact barrier potentials (or complementarity constraints) into saddle-search so that "locking" states and contact-enabled rigidity are naturally represented. [38]

3) **Surrogate modeling of energy landscapes.**
Train graph-based surrogates on motif-level simulations to predict: - number of stable states,
- approximate barrier heights,
- and sensitivity to hinge stiffness and thickness.
This aligns with the growing use of machine learning in metamaterials design but targets **energy topology**, not just final shapes or stiffness. [39]

4) **Robust design objective.**
Replace "maximize multistability" with objectives like: - maximize minimum barrier between undesired modes and desired mode,
- minimize actuation energy to reach desired state while maintaining barrier margins,
- optimize "mode selectivity" under bounded variations in thickness/hinge stiffness.

5) **Experimental validation.**
- Use quasi-static loading to map force–displacement and identify hysteresis loops.
- Use repeated cycling to see whether the same attractor is selected.
- For dynamic transitions, record high-speed kinematics and compare to predicted barrier crossings.

Difficulty, expertise, timeline.
- Difficulty: **High** for single units; **very high** for lattices with many cells and contact.
- Expertise: nonlinear dynamics, computational mechanics, optimization, statistical learning, experiments.
- Timeline: - **1–2 years**: robust atlas tools for single motifs and small multi-cell patches.
- **3–5 years**: reliable landscape surrogate models usable in design loops.
- **5–8 years**: multi-cell reprogrammable lattice landscape design with quantified robustness.

Design Synthesis Methodologies: Core Unsolved Problems

General inverse design from target shape/function to crease/cut patterns under physical and manufacturing constraints

Problem statement (a).
Given a target specification—shape trajectory, deployed surface, load-bearing/stiffness targets, or reconfiguration graph—compute a **manufacturable** crease/cut pattern (and thickness/hinge design) that achieves it, while satisfying: - foldability/deployability constraints,
- self-avoidance/contact constraints,
- thickness and hinge realizability,
- and robustness to fabrication tolerances.

This is the central “CAD for foldable matter” challenge across origami, kirigami, and hybrid metamaterials. [40]

Why it is unsolved (b): key technical challenges.
- The “inverse origami problem” has algorithmic solutions for certain target classes (e.g., polyhedral shapes), but target specifications in engineering are far richer (nondevelopable surfaces, multi-functional objectives, thickness constraints). [41]
- Search spaces are combinatorial (crease assignments, cut topology) and geometric (angles/lengths), often highly nonconvex.
- Physical constraints (contact, elasticity, buckling) make forward simulation expensive and gradients hard to obtain reliably, which slows optimization and can trap designs in local minima. [42]

Prior approaches and limitations (c).
- A practical algorithm exists to generate crease patterns that fold polyhedra (“Origamizer”-style approaches), but this does not directly address mechanical performance, thickness, and multi-physics integration. [43]
- Additive algorithms for developable quadrilateral surfaces expand design capability, but practical engineering asks for robustness, load paths, and manufacturable hinges. [44]
- Inverse design of deployable origami structures (notably for quadrilateral mesh families) uses optimization and efficient parameterizations, but remains family-specific and does not unify kirigami cuts, contact, and multi-physics. [45]
- For 2D→3D buckling-guided assembly and kirigami-based surface formation, topology optimization and analytic stiffness-based strategies exist, but scaling to arbitrary surfaces with strict manufacturing constraints remains challenging. [46]

Concrete technical routines to address the problem (d).

A plausible “next-generation” synthesis stack:

1) **Two-level representation: topology then geometry.**
- Level A (topology): represent patterns as graphs/hypergraphs (creases/cuts as edges; panels as faces).
- Level B (geometry): optimize angles, lengths, thicknesses, hinge placements.

2) **Constraint screening via rigidity theory and local foldability tests.**
Use Jacobian rank tests / local rigidity concepts to prune topologies that cannot meet intended DOF or that are overconstrained in unhelpful ways, leveraging advances in rigid origami rigidity analysis. [12]

3) **Differentiable forward model with contact and thickness hooks.**
Embed a reduced-order bar-and-hinge / shell hybrid with contact barriers (as above) into an adjoint-capable solver so gradients are available for geometry optimization. Contact simulation work makes a strong case that barrier-style contact potentials are feasible in origami contexts. [47]

4) **Multi-objective robust optimization.**
Formulate: - objective: match target shape + meet stiffness/strength,
- constraints: foldability path existence, thickness feasibility, maximum stress/strain, collision avoidance,
- robustness: worst-case or chance constraints over manufacturing tolerances.

5) **Learning-accelerated candidate generation.**
Use generative models (graph generative networks, diffusion over topologies) to propose candidate topologies that are then verified by physics-based screening. The broader metamaterials community increasingly uses machine intelligence for design; integrating it with hard foldability/contact constraints is a key open step. [39]

6) **Verification protocol.**
"Design is correct" should mean: - it folds along a validated path without penetration,
- it meets mechanical metrics in at least two independent models (reduced-order + high-fidelity localized FE),
- and it meets prototype tests within a pre-registered margin (shape error, stiffness, deployment force). [48]

Difficulty, expertise, timeline.
- Difficulty: **Very high** (because it combines NP-hard-like combinatorics with nonlinear mechanics). [49]
- Expertise: computational design, mechanics, contact simulation, optimization, ML, manufacturing.
- Timeline: - **1–3 years**: robust inverse design for restricted pattern families (common tessellations, constrained cut motifs) with contact-aware validation. [50]
- **3–6 years**: general pipelines that support both origami folding and kirigami cuts, with thickness-aware fabrication constraints.
- **6–10+ years**: near-CAD-level generality across wide applications (still likely to require family assumptions).

Inverse design of effective properties and wave/topological functionalities across reconfiguration states

Problem statement (a).
Design origami/kirigami metamaterials whose **effective properties** (stiffness tensor, Poisson response, multistage stiffness, damping, wave band structures, topological boundary modes) are programmable over configurations, while ensuring: - stable and repeatable reconfiguration,
- robustness to defects and stress history,
- and manufacturability and durability.

Why it is unsolved (b): key technical challenges.
- Property targets are often **high-dimensional** (full elasticity tensor, dispersion

relations), and must be satisfied not in one configuration but across a **configuration manifold**. [51]
- Reconfiguration by folding can itself change constraints and induce frustration/defects, complicating predictability. [52]
- Topological mechanics in origami/kirigami introduces nonlocal invariants and boundary-localized modes; controlling these via folding is promising but requires coupling geometry, mechanics, and sometimes material phase changes (e.g., shape memory). [53]

Prior approaches and limitations (c).
- Topological mechanics of origami/kirigami established design principles for localized folding motions tied to topological invariants, but translating these to robust, reconfigurable, multi-state devices under realistic imperfections remains challenging. [54]
- Lockable origami-inspired metamaterials show multi-directional stiffness and reprogrammability via contact-generated topology/symmetry changes, but robustly predicting lock conditions and property repeatability across cycles is still open. [36]
- Machine-learning-assisted design and inverse design with genetic algorithms have been demonstrated for kirigami and metamaterials, but reviews emphasize time-consuming nonlinear optimization loops and limited generalization across targets. [55]
- Kirigami engineering reviews outline deterministic geometry→mechanics relationships and methodologies, but "closing the loop" into robust inverse design for advanced metamaterial functionalities remains a frontier. [56]

Concrete technical routines to address the problem (d).

1) **Property-to-geometry via differentiable homogenization.**
Compute effective tensors from unit-cell simulations and differentiate through them (adjoint methods) to enable gradient-based optimization over cut/fold parameters.

2) **Multi-configuration constraints.**
Instead of optimizing a single state, enforce constraints at a set of sampled configurations $\{q_k\}$ and include regularizers that keep the configuration path feasible (no collisions, bounded strains).

3) **Topology-aware objectives.**
For topological modes, include objectives based on: - band inversions / gap measures,
- edge vs bulk response ratios,
- and invariants where computable, extending origami/kirigami topological mechanics concepts into optimization. [57]

4) **Stress-history modeling and "reset" strategies.**
Recent work on transformable topological metamaterials indicates that stored

elastic energy (stress history) can degrade topological response, and that material phase transitions may "shield" response ("stress caching" mitigation). Incorporate such state variables into design/validation for reconfigurable origami/kirigami as well. [58]

5) **Benchmark suite.**
Define standard target sets: - isotropic→anisotropic stiffness modulation,
- auxeticity switching,
- edge-mode transfer by folding (e.g., Kresling-based ideas),
and require designs to hit targets under both ideal and perturbed parameter sets. [59]

Difficulty, expertise, timeline.
- Difficulty: **High to very high** (physics + multi-objective + robustness).
- Expertise: metamaterials homogenization, wave mechanics, topology, optimization, ML.
- Timeline: - **1–3 years**: robust pipelines for stiffness/Poisson switching in restricted families; more reliable surrogate generalization. [60]
- **3–7 years**: mature design loops for wave/topological functionalities with experimental validation. [57]

Multi-Physics Implementation, Control, and Sensing: Core Unsolved Problems

Co-design of actuation, sensing, and control for high-DOF foldable robots and "field-programmable" sheets

Problem statement (a).
Create methods to design foldable robots/material systems in which: - actuation (thermal, pneumatic/vacuum, magnetic, etc.), - sensing (fold angle, strain, temperature, contact state), - and control (closed-loop policies) are **co-designed with the structure**, enabling reliable reconfiguration across tasks and environments without refabrication.

Why it is unsolved (b): key technical challenges.
- Foldable systems often have **many DOFs** with strong coupling, friction/contact events, and hysteresis; state estimation is difficult without distributed sensing. [61]
- Theoretical branch/mode multiplicity near flat states implies that actuation must be carefully placed and sequenced to avoid undesired branches (a fundamental limit for self-folding). [14]
- Hardware constraints (heater density, wiring, pneumatic routing, magnetic embedding) impose design constraints that most geometry-only synthesis tools ignore.

Prior approaches and limitations (c).
- A field-programmable robotic folding sheet demonstrates dense electro-thermo-

responsive elements that serve as both heaters and sensors, with closed-loop control enabling real-time fold programming post-deployment. This is a major step, but generalizing to other actuation modalities and guaranteeing performance under contact-rich tasks remains open. [62]
- Modular origami robots with electrothermal actuation show plug-and-play assembly of multi-DOF systems, but integrated sensing/control co-design remains an active frontier. [63]
- Model-based control of proprioceptive origami actuators for manipulators has been proposed, illustrating progress, but scaling to broader pattern classes and dynamic terrain interaction remains challenging. [64]
- Reviews of kirigami robots stress that kirigami can simplify robot structures by integrating actuation/sensing/skin functions, but this vision needs systematic co-design methods and standardized control abstractions. [65]

Concrete technical routines to address the problem (d).

1) **Structure–electronics co-layout optimization.**
Introduce a joint design variable set: - geometric pattern G (creases/cuts),
- actuator layout A (heater traces, pneumatic channels, magnet positions),
- sensor layout S (resistive/thermal/strain sensors),
and solve a multi-objective optimization: - controllability/observability metrics (linearized around configurations),
- energy consumption and thermal constraints,
- achievable fold angles and speed,
- robustness to branch ambiguity.

2) **Hybrid dynamical models for control.**
Model as a hybrid system: - continuous dynamics within a mode,
- discrete transitions at contact/locking or snap-through events, which are essential in multistable origami/kirigami. [66]

3) **System identification loop.**
Use closed-loop excitation to identify: - hinge stiffness and damping,
- actuation transfer functions (electrothermal time constants, pneumatic compliance),
- friction/contact thresholds.
Field-programmable sheets motivate using embedded sensing for self-calibration. [67]

4) **Contact-aware control policies.**
Incorporate contact predictions from barrier models into MPC or sampling-based control: - constraints enforce nonpenetration,
- terminal conditions capture “lock achieved” states for structural functions (deployable stiffness). [38]

5) **Validation protocol.**
- Closed-loop performance: rise time, overshoot, final shape error across multiple targets.
- Repeatability: success rate across repeated folds and environmental variations (temperature, humidity).
- Safety: maximum stresses/temperatures vs material limits.

Difficulty, expertise, timeline.
- Difficulty: **High** (hardware + nonlinear hybrid control).
- Expertise: robotics, embedded systems, thermal/fluid/EM actuation, state estimation, hybrid control, mechanics.
- Timeline: - **1–3 years**: co-design frameworks for specific actuation modes (electrothermal, pneumatic) with open benchmark tasks, building on demonstrated field-programmable architectures. [68]
- **3–6 years**: generalized abstractions ("fold primitives," mode graphs) that transfer across designs and enable libraries of behaviors.
- **6–10 years**: widespread field-deployable foldable robotic materials with strong reliability guarantees.

Practical Applications and Translation: Core Unsolved Problems

Durability, fatigue, and environmental robustness of creases/hinges and integrated multi-material systems

Problem statement (a).
Predict and extend the **lifetime** of origami/kirigami mechanisms, especially where: - deformation localizes in hinges/creases,
- multi-material interfaces (electronics on flexible substrates, fiber reinforcements, adhesives) experience cyclic strain, and
- environments (humidity, temperature, UV, chemical exposure) alter properties.

Why it is unsolved (b): key technical challenges.
- Creases/hinges combine geometric nonlinearity, material nonlinearity, and sometimes damage accumulation.
- Failure modes include plastic hinge fatigue, delamination, crack initiation at cut tips, and drift in actuator response (e.g., electrothermal aging).
- There is no universally adopted lifecycle test standard tailored to foldable mechanisms.

Prior approaches and limitations (c).
- Work on origami electronics explicitly highlights hinge durability challenges and proposes reinforcement strategies (e.g., fiber-reinforced origami electronics) but broader generalized lifetime prediction remains open. [69]
- An ultra-low cycle fatigue plastic hinge line model for origami structures has been proposed and validated on a waterbomb tubular origami, demonstrating progress

toward damage models; however, generalization across materials and folding patterns is still early. [70]
- Modular and thick-panel origami research demonstrates load-bearing feasibility at larger scales, but does not by itself solve long-term fatigue and environmental cycling. [71]

Concrete technical routines to address the problem (d).

1) **Standardized hinge coupon taxonomy.**
Define a small set of hinge archetypes: - living hinges (polymer),
- laminated flexures,
- composite-reinforced hinges,
- printed multi-material hinges,
and require reporting of geometry, thickness, radius, and measured moment–rotation curves over cycles. [72]

2) **Physics-based damage models + parameter ID.**
Adopt a damage variable $D \in [0,1]$ localized to creases/hinges (as in plastic hinge-line approaches) and identify parameters from cycle tests, then propagate into full-structure simulations. [73]

3) **Fracture-aware kirigami design rules.**
Develop cut-tip stress intensity metrics and use them as constraints in inverse design; incorporate rounding/fillets and hierarchical cut patterns where appropriate (hierarchical kirigami studies provide geometric modeling pathways). [74]

4) **Environmental accelerated testing.**
- humidity/temperature cycling,
- UV exposure,
- corrosion/chemical exposure (as relevant),
paired with periodic calibration of stiffness/actuation. Early origami robotic demonstrations already note sensitivity to humidity and thickness in self-folding contexts, underlining the need for environmental protocols. [75]

5) **Reliability metrics for reporting.**
Report: - cycles to stiffness loss threshold,
- cycles to fold-angle error threshold,
- probability of correct mode selection under repeated actuation,
- and drift in actuation energy.

Difficulty, expertise, timeline.
- Difficulty: **Moderate to high** (experiments heavy but conceptually tractable).
- Expertise: materials & fatigue, experimental mechanics, reliability engineering, modeling.
- Timeline: - **1–2 years**: community hinge coupon standards + shared datasets.

- **3–5 years**: widely adopted damage models integrated into design tools for major families.

Scalable fabrication, micro-to-macro process-aware design, and standardized validation/benchmarks

Problem statement (a).
Build a translation pipeline that supports: - scalable manufacturing (laser cutting, roll-to-roll patterning, multi-material printing, microfabrication), - predictable assembly (including mechanically guided 2D→3D transformations), - and standardized benchmarks so that designs and algorithms can be compared across labs and applications.

Why it is unsolved (b): key technical challenges.
- At micro scales, material thickness and residual stresses dominate; at macro scales, gravity, friction, and large-load requirements dominate. Mechanically guided assembly demonstrates broad feasibility but requires careful stiffness patterning and process control. [8]
- Patents indicate application pull (deployables, medical robots, energy harvesting), but academic designs often lack standardized test/verification procedures and process-aware constraints. [76]
- Many inverse design approaches remain simulation-driven with limited calibration against manufacturing variability; ML approaches can reduce computation but depend on training distributions and reproducibility. [77]

Prior approaches and limitations (c).
- Mechanically driven kirigami assembly of 3D mesostructures from 2D precursors is well established in high-impact literature, with follow-on work and patents describing deterministic assembly of complex 3D forms; however, design automation that includes manufacturing constraints and error budgets remains incomplete. [78]
- Inverse design methods for buckling-guided assembly and kirigami-based surface formation exist (topology optimization, stiffness mapping, two-step frameworks), but the field lacks standardized benchmarks and validation requirements. [79]
- Reviews call out broad applications and future challenges, but benchmarking remains fragmented (different geometries, materials, metrics). [80]

Concrete technical routines to address the problem (d).

1) **Process-aware design constraints.**
Embed fabrication constraints directly in synthesis: - minimum cut width / kerf compensation,
- minimum hinge radius,
- maximum allowable residual stress,
- layer stack constraints for electronics,
- allowable thickness variations.

2) **Cross-scale dimensionless design maps.**
Develop dimensionless groups that govern behavior shifts, e.g.: - thickness-to-panel-length ratio,
- bendability numbers (bending vs stretching dominance),
- actuation-to-stiffness ratios,
so that micro and macro designs can share scaling laws.

3) **Open benchmark suites.**
Create benchmark tasks with reference geometries and metrics (analogous to robotics benchmarks), such as: - **deploy-to-surface**: achieve target 3D surface within tolerance;
- **lock-and-bear-load**: achieve a lockable stiff state and meet load/deflection limits;
- **reconfigure-under-control**: perform N target folds with closed-loop error < ε;
- **fatigue challenge**: survive M cycles with bounded degradation.
Large-scale thick origami systems and lockable load-bearing metamaterials provide candidate benchmark archetypes. [81]

4) **Data standards for learning-based design.**
Given the growth of ML in metamaterials design, define minimal reporting: - dataset generation protocol (simulation and experimental),
- uncertainty tags,
- train/test splits by geometry families,
- and "out-of-distribution" performance reporting. [39]

5) **Patent-to-practice mapping.**
Use patents as "requirements documents": - e.g., deterministic 2D→3D assembly patents (manufacturing constraints),
- biomedical origami robot patents (biocompatibility, deployment reliability),
- kirigami-based energy harvesting patents (fatigue and bonding). [76]

Difficulty, expertise, timeline.
- Difficulty: **Moderate** for benchmarks/standards; **high** for full process-aware CAD pipelines.
- Expertise: design automation, manufacturing, metrology, experimental validation, software engineering.
- Timeline: - **1–2 years**: benchmark definitions, shared datasets, baseline solvers.
- **3–5 years**: process-aware inverse design integrated with common fabrication routes.
- **5–10 years**: robust, cross-scale "foldable mechanism CAD" with certification-like workflows.

Comparative Tables and Roadmaps

Comparison across dimensions

Unsolved problem	Domain emphasis	Discrete vs continuum	Theoretical vs practical weight	Short-term solvability	Long-term potential	Resource intensity	Representative anchors
Configuration-space & fold-path planning with contact/modes	Theory → tools	Hybrid	Strongly theoretical + algorithmic	Medium (restricted families)	High (but limits remain)	High (algorithms + simulation + experiments)	NP-hard rigid foldability; contact simulation; rigidity analysis [82]
Thickness-aware unified mechanics + reduced-order error control	Theory ↔ practice	Hybrid	Balanced	Medium–high	High	High (multi-scale modeling + calibration)	Thick-panel load-bearing; modular thick origami simulations [83]
Energy landscape prediction & multistability programming	Theory ↔ design	Hybrid	Balanced	Medium (small systems)	High	High (nonlinear dynamics + experiments)	NEB folding paths; lockable stiff states; branch complexity [84]
Inverse design under contact/thickness/manufacturing	Design synthesis	Hybrid	Design-heavy but theory-	Medium (family-	Very high	Very high	Origamizer; additive design;

Unsolved problem	Domain emphasis	Discrete vs continuum	Theoretical vs practical weight	Short-term solvability	Long-term potential	Resource intensity	Representative anchors
constraints			- limited	specific)			inverse design optimization; buckling assembly [85]
Inverse design of effective properties & topological/wave functions	Design synthesis	Often lattice/continuum-hybrid	Strongly theoretical + experimental	Medium	High	High	Topological mechanics; ML metamaterials design; kirigami engineering review [86]
Co-design of actuation/sensing/control for foldable systems	Practice ↔ robotics	Hybrid	Strongly practical	Medium	High	Very high (hardware + control)	Field-programmable folding sheet; modular origami robots; model-based control [87]
Durability & fatigue models for	Practice	Both	Strongly practic	High	High	Medium–high (experi	Hinge durability in

Unsolved problem	Domain emphasis	Discrete vs continuum	Theoretical vs practical weight	Short-term solvability	Long-term potential	Resource intensity	Representative anchors
creases/hinges/electronics			al			ments)	origami electronics; fatigue hinge-line model [88]
Scalable fabrication + benchmarks/standards (micro→macro)	Practice → ecosystem	Both	Strongly practical	High (benchmarks)	Very high	Medium (benchmarks) to high (CAD pipelines)	Mechanically guided assembly; patents; kirigami inverse design [89]

Research-gap intensity chart

Qualitative "gap intensity" (■ = higher unmet need relative to current methods), synthesized from the challenges repeatedly emphasized in theory, simulation, and application-focused works. [90]

- Configuration-space + contact + branching: ■■■■■
- General inverse design with manufacturability: ■■■■■
- Energy landscape + robust multistability: ■■■■□
- Thickness-aware unified mechanics w/ uncertainty: ■■■■□
- Co-design actuation/sensing/control: ■■■■□
- Property/topology programmable metamaterials: ■■■□□
- Durability/fatigue standards and models: ■■■□□
- Fabrication benchmarks and shared datasets: ■■□□□

Dependency roadmap

```
flowchart TD
  A[Contact-aware forward simulation] --> B[Fold-path planning & mode
```

```
graphs]
  A --> C[Energy landscape atlas]
  D[Thickness-aware mechanics models] --> A
  D --> E[Process-aware inverse design constraints]
  B --> F[General inverse design (shape + function)]
  C --> F
  E --> F
  F --> G[Co-design actuation + sensing + control]
  G --> H[Application deployment + benchmarking]
  H --> D
```

Plausible progress timeline

```
gantt
  title Plausible timelines for progress (relative, not guaranteed)
  dateFormat  YYYY
  axisFormat  %Y
  section Near-term (1–3 years)
  Benchmarks & shared datasets           :a1, 2026, 2y
  Restricted-family contact+path tools   :a2, 2026, 3y
  Hinge coupon standards + fatigue IDs   :a3, 2026, 2y
  section Mid-term (3–6 years)
  Process-aware inverse design pipelines :b1, 2027, 4y
  Landscape surrogates for motifs        :b2, 2027, 4y
  Co-design toolchains for key actuators :b3, 2027, 4y
  section Long-term (6–10+ years)
  Broadly general foldable CAD w/ certif :c1, 2029, 6y
  Topology/wave/state programmable systems: c2, 2029, 6y
```

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